

Docker + GitHub Template Cheat Sheet

Use this as a lab template. Replace YOUR_PROJECT_NAME, YOUR_CONTAINER_NAME, and YOUR_USERNAME with your actual values.

1. Docker Commands

Check Docker version

```
docker --version
```

List all containers

```
docker ps -a
```

List Docker images

```
docker images
```

Login to Docker Hub

```
docker login
```

Build project image

```
docker build -t YOUR_PROJECT_NAME .
```

Run the container

```
docker run -d -p 8080:8080 --name YOUR_CONTAINER_NAME YOUR_PROJECT_NAME
```

Tag image for Docker Hub

```
docker tag YOUR_PROJECT_NAME YOUR_USERNAME/YOUR_PROJECT_NAME:latest
```

Push image to Docker Hub

```
docker push YOUR_USERNAME/YOUR_PROJECT_NAME:latest
```

2. Complete Docker Template — Run in Order

```
docker --version
docker ps -a
docker images
docker login
docker build -t YOUR_PROJECT_NAME .
docker run -d -p 8080:8080 --name YOUR_CONTAINER_NAME YOUR_PROJECT_NAME
docker tag YOUR_PROJECT_NAME YOUR_USERNAME/YOUR_PROJECT_NAME:latest
docker push YOUR_USERNAME/YOUR_PROJECT_NAME:latest
```

3. GitHub Commands

Initialize Git

```
git init
```

Stage all files

```
git add .
```

Commit files

```
git commit -m "Initial commit"
```

Set main branch

```
git branch -M main
```

Connect to GitHub

```
git remote add origin https://github.com/YOUR_USERNAME/YOUR_PROJECT_NAME.git
```

Push to GitHub

```
git push -u origin main
```

4. Complete GitHub Template — Run in Order

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/YOUR_PROJECT_NAME.git
git push -u origin main
```

5. Quick Docker Troubleshooting

Task	Command
See running containers	docker ps
See all containers	docker ps -a
See images	docker images
View container logs	docker logs YOUR_CONTAINER_NAME
Stop container	docker stop YOUR_CONTAINER_NAME
Start container	docker start YOUR_CONTAINER_NAME
Remove container	docker rm -f YOUR_CONTAINER_NAME

Example: If your project is called MyHMS and your Docker Hub username is student123, replace YOUR_PROJECT_NAME with MyHMS and YOUR_USERNAME with student123.

Important: Create the GitHub repository with the same project name before running git push. Also make sure the Docker Hub repository/name matches the tag you push.